

Effect of the Activation of A2A Receptors on the Modulation of NURR1

Abstract

Twenty five percent of Americans over the age of 65 are afflicted by osteoarthritis, a painful inflammatory joint disease. A crucial regulator of inflammation is adenosine, the A_{2A} receptors specifically are profoundly responsive to changes in exogenous and endogenous factors involved in inflammation. A receptor protein, NURR1, is thought to have anti-catabolic and anti-inflammatory effects within cartilage. By activating A2A receptors in chondrocytes we expect NURR1 levels will increase. This was studied by treating TC28a2 chondrocyte cells with A2A agonists and antagonists and then measuring NURR1 levels after different treatments. The results suggested that when A2A receptors were activated NURR1 levels increased and when A2A receptors were blocked NURR1 levels remained at the standard. Due to NURR1's anti-catabolic and inflammatory effects in chondrocyte tissues this could lead to a better understanding of OA and its treatment.

Introduction

Osteoarthritis

Osteoarthritis (OA) is the most common type of arthritis. Twenty five percent of the US population above the age of 65, and millions more worldwide, suffer as a result of OA¹. Osteoarthritis is characterized by the degeneration of joint cartilage and the underlying bone^{2,3}. The symptoms that characterize OA are primarily pain and stiffness. Osteoarthritis can affect all joints, but in most cases it affects the hip, knee, and thumb joints, and the spine. Chondrocytes, the cells that secrete and maintain cartilage, play a key role in driving OA.⁴ In response to mechanical overloading, chondrocytes release proteolytic enzymes and inflammatory mediators, which cause more damage to the cartilage. When cartilage in joints is damaged the underlying bone is affected, causing more inflammation.⁵

The pathophysiology of osteoarthritis includes cartilage destruction, synovial inflammation, and osteophyte formation.⁶ OA pathophysiology often begins as a result of injury

¹Corciulo, Carmen, et al. "Endogenous Adenosine Maintains Cartilage Homeostasis and Exogenous Adenosine Inhibits Osteoarthritis Progression." *Nature Communications*, vol. 8, 2017, p. 15019., doi:10.1038/ncomms15019.

²Lieberthal, J., Sambamurthy, N. & Scanzello, C. R. Inflammation in joint injury and post- traumatic osteoarthritis. *Osteoarthritis Cartilage* 23, 1825–1834 (2015).

³Brown, C., Fisher, J. & Ingham, E. Biological effects of clinically relevant wear particles from metal-on-metal hip prostheses. *Proc. Inst. Mech. Eng. [H]*. 220, 355–369 (2006).

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to the cartilage.⁷ Cartilage can be damaged as a result of trauma or as a result of general wear and tear. Injury frequently disrupts the homeostasis of the cartilage, which is what then drives OA progression.⁸ Cartilage is a distinctive tissue with viscoelastic and compressive characteristics that result from its extracellular matrix consisting of proteoglycans and type two collagen.⁹ In healthy cartilage tissue this matrix is constantly undergoing a dynamic process of remodelling, regulated by low levels of degradative and synthetic enzymes that work in tandem to maintain cartilage volume.¹⁰ However, in OA, whether driven by trauma or other factors, this balance is disrupted and matrix degrading enzymes become overexpressed.¹¹ The enzymes that are thought to mediate cartilage destruction are matrix metalloproteinases (MMPs) and proinflammatory cytokines (e.g., IL-1).¹² When the cartilage degrades, the joint space narrows, leaving the underlying bone vulnerable to damage. Without the protection of cartilage, underlying bones begin to change, forming smooth bony growths or deposits called osteophytes. As shown in Figure 1, OA progression is generally characterized by four stages, with one being the least severe and 4 being the most severe. What begins as a small disruption in the delicate homeostasis of cartilage matrix maintenance can progress to debilitating pain and bone destruction.¹³

⁷ "Osteoarthritis - Pathophysiology - Clinical Features." TeachMeSurgery. Accessed February 20, 2019. <http://teachmesurgery.com/orthopaedic/principles/osteoarthritis/>.

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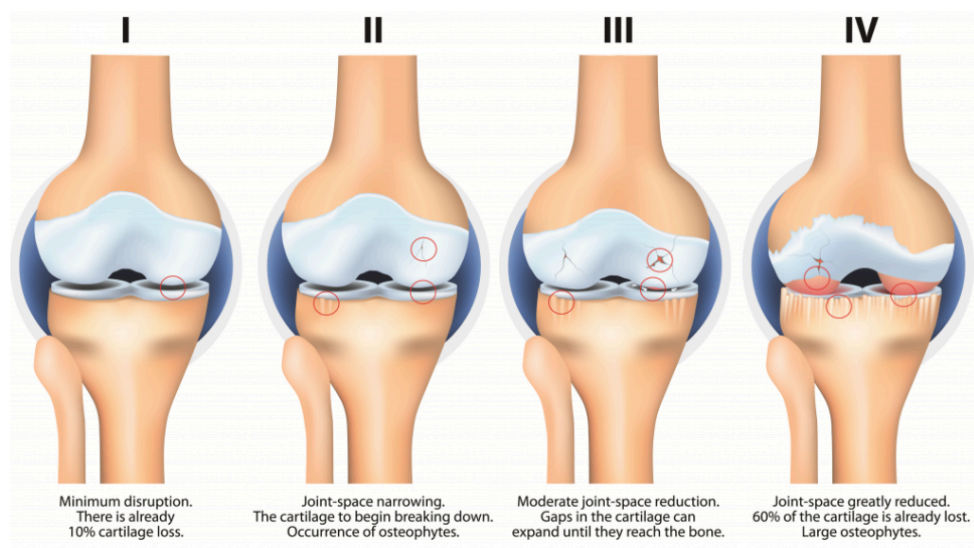


Figure 1 – The stages of osteoarthritis, as demonstrated in the knee joint

Figure 1. The Stages of Osteoarthritis, as Demonstrated in the Knee Joint. Shown are the four stages of osteoarthritis progression. Stage I is characterized by 10% cartilage loss. In Stage II, as cartilage breaks down, the joint-space narrows and osteophytes begin to form. Stage III is characterized by continued joint-space reduction and gaps in cartilage. By Stage IV, 60% of cartilage has been lost and large osteophytes are present.¹⁴

Adenosine

Inflammation plays the central role in the development of OA¹⁵. A crucial regulator of inflammation is adenosine. Adenosine is a purine nucleoside often produced in response to cell damage and metabolic stress.¹⁶ Adenosine levels are elevated in conditions such as hypoxia, ischemia, trauma, and inflammation. The primary pathway that yields elevated extracellular adenosine levels as a result of metabolic stress is the release of precursor adenine nucleotides (predominantly ATP) from cells, followed by extracellular catabolism of adenosine through a cascade of ectonucleotidases. Ectonucleotidases are membrane enzymes that metabolize nucleotides to nucleosides.¹⁷ In order to produce adenosine, ATP/ADP must first be converted to AMP. This conversion is carried out by ENTPD1, which is also known as CD39. The next step is carried out by NT5E, which is also known as CD73, and converts AMP into adenosine. High

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¹⁵ Abramson, S. B. & Attur, M. Developments in the scientific understanding of osteoarthritis. *Arthritis Res. Ther.* 11, 227 (2009).

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extracellular adenosine levels can also result from the release of intracellular adenosine, which occurs when ATP is degraded in ischemic conditions.¹⁸

Adenosine induces a broad array of physiological responses. Extracellular adenosine mediates these effects by activating one or more of four G-protein-coupled adenosine receptors: A₁, A_{2A}, A_{2B}, and A₃. A_{2A} receptors are universal within the body.¹⁹ The gene for A_{2A} receptors is comprised of multiple exons that encode alternative transcripts, the expression of which is driven by a minimum of four independent promoters. Studies on the regulation of these promoters have proved that A_{2A} receptor gene expression is heavily affected by changes in the extracellular environment.²⁰ A_{2A} receptor expression is profoundly responsive to changes in concentrations of exogenous and endogenous factors that are involved in inflammation. This suggests that A_{2A} is the key adenosine receptor to target.²¹ In studies examining the role of A_{2A} receptors it is vital to be able to inhibit or activate A_{2A}.²² Two prominent drugs that can be used to control the A_{2A} receptor are the antagonist ZM241385, which blocks adenosine from binding to and activating the receptor, and the agonist CGS21680, which constitutively activates the A_{2A} receptor.²³

NURR1

NURR1, along with NOR-1 and NURR77, compose the NURR subfamily of orphan receptors within the nuclear receptor superfamily.²⁴ Orphan receptors are receptors whose endogenous ligand is unknown. Nuclear receptors are a category of intracellular proteins responsible for sensing steroid and thyroid hormones, among other molecules.²⁵ In response, nuclear receptors work alone or with other proteins to regulate specific gene expression. Thus, these proteins control the homeostasis, metabolism, and development of an organism. NURR subfamily receptors are products of immediate early genes, meaning that their expression can be induced by a range of different extracellular stimuli.²⁶

Studies of the NURR subfamily in past joint research has suggested they play a vital roles in joint tissues.²⁷ The synovium is a fragile tissue lining the joint capsule. In the synovium low doses of methotrexate, a drug used to treat certain cancers and rheumatoid arthritis, lowered NURR1 levels, leading to a decrease in Synovial inflammation.²⁸ This process was found to be mediated by A_{2A} receptors. Osteoblasts (cell that secrete bone formation matrix), NURR1 was found to regulate differentiation.²⁹ In Osteoclasts (cell that breaks down bone tissue), NURR1

¹⁸ Haskó, György. Pacher, Pál. "A2A receptors in inflammation and injury: lessons learned from transgenic animals." NIH Public Access. .

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²⁴ Murphy, Evelyn, et al. "MODulation of Orphan Nuclear Receptor NURR1 Expression by Methotrexate in Human Inflammatory Joint Disease Involves Adenosine A2A Receptor-Mediated Responses" The Journal of Immunology.

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levels decreased during osteoclast differentiation. This suggests that increasing NURR1 levels within bones would encourage bone formation and inhibit bone degradation.³⁰ In a comparison between chondrocytes (cartilage cells) with and without osteoarthritis, levels of NURR77, NURR1, and NOR-1 were looked at.³¹ Of the three, NURR1 was the protein to show the largest difference, increasing drastically in response to OA.³² The NURR1 in chondrocytes is strongly activated by interleukin-1 β (IL-1 β). IL-1 β is an inflammatory mediator that contributes to OA by diminishing ATP and adenosine release from cultured chondrocytes.³³ Figure 1 suggests that NURR1 is produced in response to IL-1 β and as a result should have Anti-inflammatory and anti-catabolic effects within chondrocytes.³⁴ From this point it is not a stretch to think NURR1 has the potential to treat OA in the cartilage.³⁵

With the potential of NURR1 to lower inflammation the next goal is to figure out how to modulate NURR1 in the chondrocytes. Based on trends in previous research, A_{2A} receptors are a promising candidate for exploring the regulation of NURR1 levels.³⁶ By treating cells with the A_{2A} agonist CGS and antagonist ZM we can begin to understand how and if A_{2A} modulates NURR1 in chondrocytes.

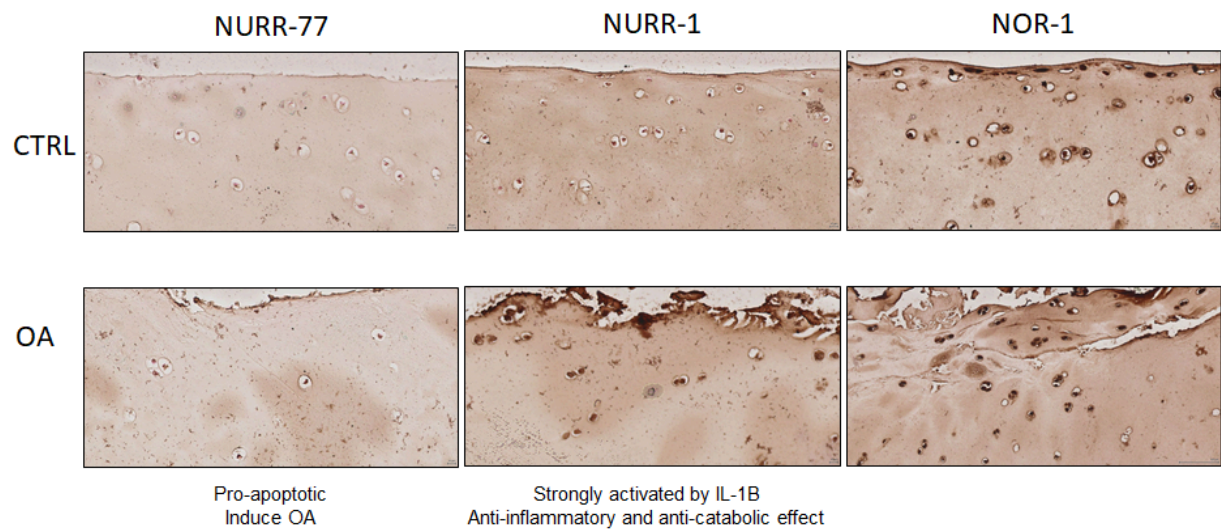


Figure 2. Chondrocyte Cells Stained for Different NURR Subfamily Proteins in Healthy and Osteoarthritic Cells. Pictured above are cartilage cells stained for NURR-77, NURR1, and NOR-1 presence in control and osteoarthritic cells.³⁷

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³¹ Corciulo, Carmen. Personal Communication. August 2018 .

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Materials and Methods

Human chondrocyte TC28a2 cells were cultured in MEM α media at 37°C in 50 mL flasks. The cells were grown until the flasks reached 70% coverage, at which point they were split. Once a sufficient amount of cells were cultured, 500,000 cells were plated per 10 mL petri dish. The petri dishes were incubated at 37°C for 48 hours. The cells were then treated with ZM241385 (4-{2-[7-amino-2-(2-furyl) (1, 2, 4)triazolo (2,3-a) (1, 3, 5)triazin-5-yl-amino]ethyl}-phenol) (an A_{2A} antagonist) and CGS21680 (2-[p-(2-carboxyethyl)phenylethyl-amino]-5'-N-ethyefresh media was added to each petri dish. They were then incubated at 37°C for their respective times.

To remove the drug, cells were washed twice with PBS. Then cells were collected and prepared for protein extraction. The nuclear and cytosolic proteins were isolated. Next, conducting colorimetric microplate protein quantification assays were used to determine the concentration of proteins in each sample. Western blots were then run to determine the amount of NURR1 in the cytosol and nucleus of the cells that were treated with either CGS or ZM for different time periods. After gel electrophoresis, the western blots were blotted onto membranes. The gel is placed on a membrane and the two are then sandwiched between two buffer soaked filter papers. An electric current is run through membrane and gel for two hours. The membranes were treated with Fetal Bovine Serum on a shaker for 60 minutes in order to “cover” background proteins. After being washed with Tris Buffered Saline Tween 20 the membranes are then treated with NURR1 and GAPDH antibodies overnight on a shaker at -20° C. Finally they were treated with HRP, a fluorescent antibody, and scanned using a CCD imaging device.

Results

As shown in Figure 3 and 4, TC28a2 chondrocyte cells were treated with either CGS or ZM and then the amount of NURR1 in the cytosol (Figure 3) or nuclei (Figure 4) of the cells was measured. All protein levels were normalized using GAPDH protein levels. Shown is the fold change in protein relative to the untreated cells, which were set to 1. In the cytosol (see Figure 3), when cells were treated with CGS, NURR1 levels rose to 1.4 fold change with five minutes of CGS treatment and 1.5 with fifteen minutes of treatment. After this point, NURR1 levels began to lower, reaching 1.1 fold change at thirty minutes, and just over 0.8 after an hour. Between fifteen and thirty minutes NURR1 levels decreased more rapidly than between thirty minutes and an hour. When treated with ZM, chondrocyte cytosol NURR1 levels had an opposite trajectory compared to when treated with CGS. The fold change in NURR1 levels was reduced at five minutes and there was no fold change at ten minutes. However, when treated with ZM for thirty minutes, NURR1 levels reached a 1.2 fold change, which was maintained at an hour. In the nucleus (see Figure 4), NURR1 levels increased rapidly in the nuclei after five minutes of CGS treatment and peaked at 3 fold change with fifteen minutes of treatment. With 30 minutes of CGS treatment NURR1 levels went down to about 0.75 fold change and increased only slightly

with an hour of treatment. Nucleic NURR1 levels decreased after five minutes of ZM treatment to 0.75 fold change, and NURR1 continued to decrease until 30 minutes of treatment, after which NURR1 increased slightly after an hour.

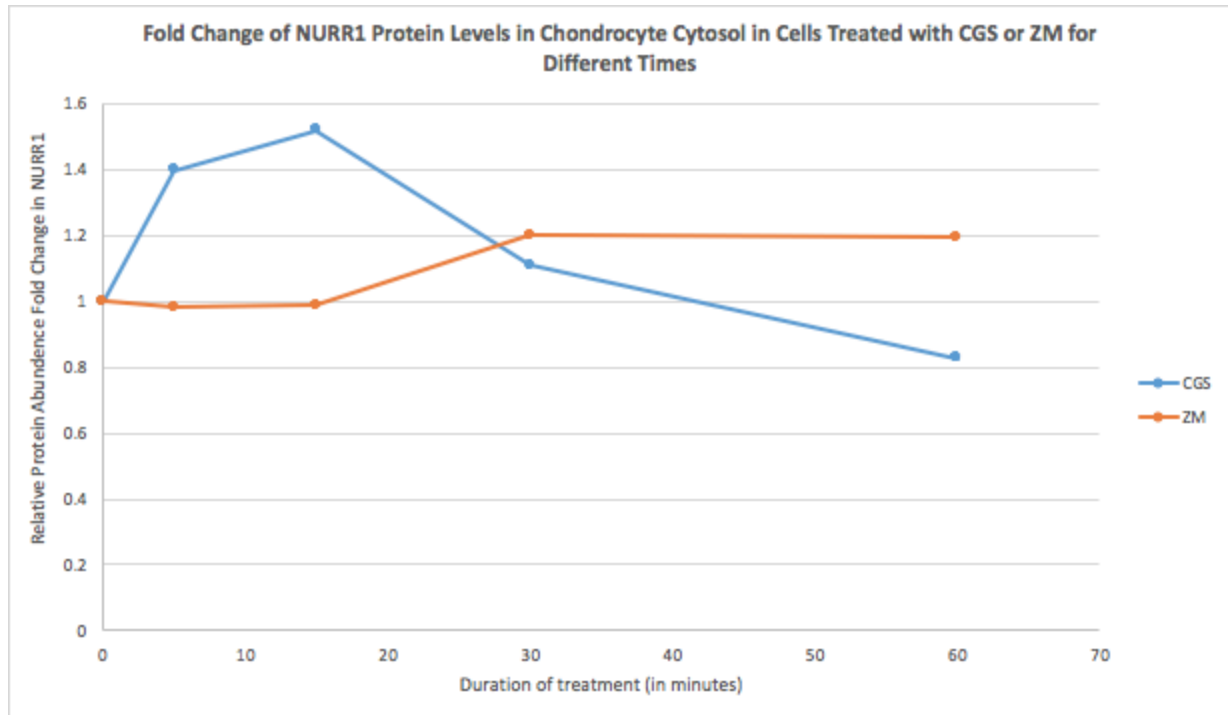


Figure 3. Fold Change of NURR1 Protein Levels in Chondrocyte Cytosol in Cells Treated with CGS or ZM for Different Times. TC28a2 chondrocyte cells were treated with either CGS or ZM and then the amount of NURR1 in the cytosol of the cells was measured. All protein levels were normalized using GAPDH protein levels. Shown is the fold change in protein relative to the untreated cells, which were set to 1.

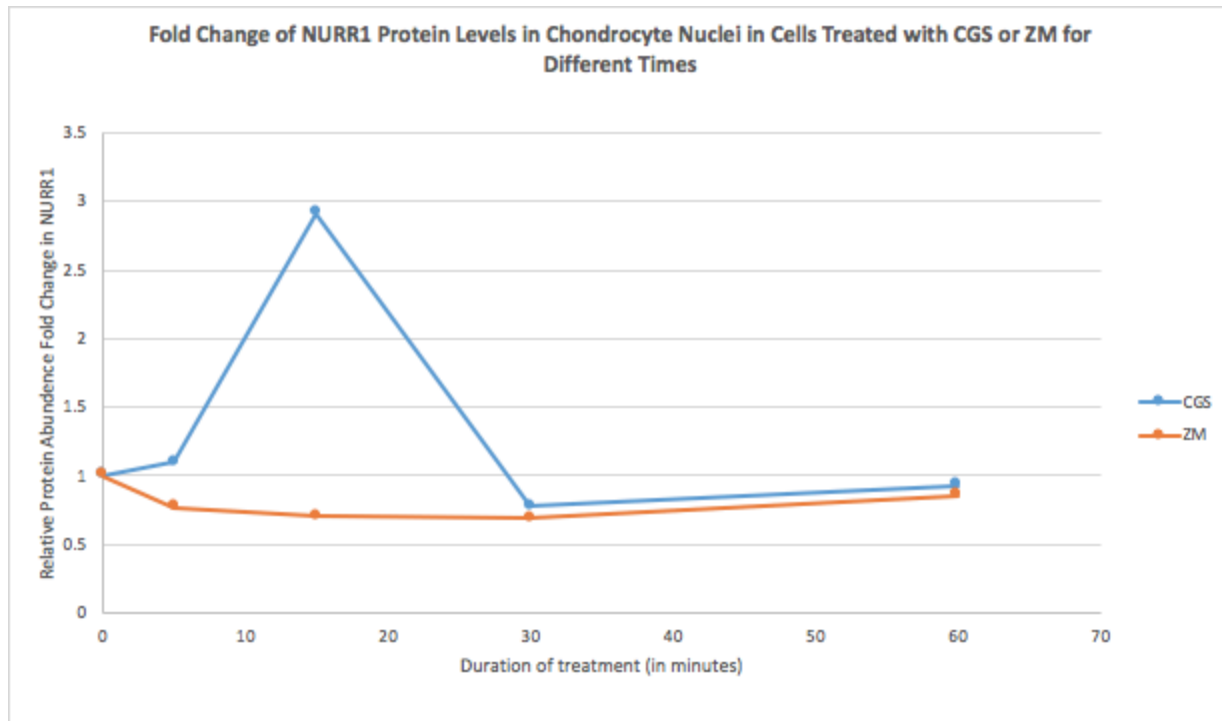


Figure 4. Percentage of NURR1 in Chondrocyte Nuclei in Cells Treated with CGS or ZM for Different Times TC28a2 chondrocyte cells were treated with either CGS or ZM and then the amount of NURR1 in the nuclei of the cells was measured. All protein levels were normalized using GADPH protein levels. Shown is the fold change in protein relative to the untreated cells, which were set to 1.

Discussion

Osteoarthritis is a painful inflammatory joint disease that afflicts 25% of Americans over the age of 65 and millions more worldwide. Adenosine is a crucial regulator of inflammation, and A_{2A} receptors are profoundly responsive to changes in concentrations of exogenous and endogenous factors that are involved in inflammation. NURR1 is a receptor protein thought to have anti-inflammatory and anti-catabolic effects within chondrocytes.

As seen in Figure 3, when A_{2A} is activated by the agonist CGS, cytosolic NURR1 levels rise for the first 15 minutes of treatment, then fall to just below the standard NURR1 levels with 60 minutes of treatment. It is possible that the initial accumulation of NURR1 may be due to the protein being synthesized more quickly than it can be transported into the nucleus. As the synthesis of NURR1 slows down and the accumulated NURR1 is transported into nucleus, cytosolic levels of NURR1 drop, as seen in Figure 3. Figure 4 suggests that A_{2A} activation leads to a rise in NURR1 levels in the nucleus, gradually with five minutes of treatment, and rapidly with fifteen minutes of treatment, reaching a 3 fold change in NURR1. After 30 minutes of treatment NURR1 levels fall back close to the standard. It is possible that NURR1 levels rise slowly at first because only some NURR1 can travel into the nucleus in the limited period of

time. The rapid increase seen after fifteen minutes is likely due to the abundance of NURR1 synthesized in the cytosol travelling into the nucleus. After this the fall of NURR1 levels to standard can potentially be attributed to the NURR1 having carried out its effects and the protein degrading. As seen in both Figure 3 and 4, when A_{2A} is blocked with the antagonist ZM, NURR1 levels remain close to the standard. This implies that the A_{2A} receptor is solely responsible for activating the synthesis of or directly synthesizing NURR1.

This experiment suggests that NURR1 levels can be modulated through the activation or blocking of A_{2A} receptors. This is important because NURR1, having anti-catabolic and inflammatory effects in cartilage tissues, is likely has the ability to slow down or prevent osteoarthritis progression. If NURR1 levels can be raised there is potential for a non surgical OA treatment which would be life changing for the millions of people suffering from Osteoarthritis.

The primary limitation on this experiment is that due to time constraints it was only executed once. For the results to carry weight and be more conclusive it should have been repeated at least two more times. The observed maximum effect of the A_{2A} agonist and antagonist was with 15 minutes of treatment, assuming this finding was maintained after 2 more trials, observing NURR1 levels at 1 minute intervals from 10-20 minutes would reveal more details about the effects A_{2A} activation has on NURR1 in chondrocytes. The next limitation is that this experiment was conducted in cell culture, not *in vivo*, and the effects of the drugs might vary as a result. In the future the effects of A_{2A} activation on NURR1 levels and later joint health could be understood through trials in mice or rats and eventually humans. This experiment is just the beginning of understanding how to modulate NURR1 in chondrocytes and with further study there is potential for an osteoarthritis treatment.

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Acknowledgements

I would like to acknowledge Carmen Corciulo and Bruce Cronstein for mentoring me through the research process. I would also like to acknowledge Anne Kloimwieder for advising me through writing this paper.